FNCE30007 DERIVATIVE SECURITIES
Tutorial 5 Solutions

1. Consider the one period binomial model for a stock price. Using the following results presented in lectures:

$$\Delta u S_0 - f_u = \pi$$

$$\Delta S_0 - f = e^{-rT}\pi$$

Where f_u, f_d and f are the option values in the up state, down state and t=0 respectively, and $\Delta = \frac{f_u - f_d}{S_0(u-d)}$.
Show that: $f = e^{-r\Delta t}[p * f_u + (1-p)f_d]$
where $p = \frac{e^{r\Delta t} - d}{u-d}$

Answer:

$$\Delta u S_0 - f_u = \pi$$

$$\Delta S_0 - f = e^{-rT}\pi$$

$e^{-rT}(\Delta u S_0 - f_u) = \Delta S_0 - f$

Given $\Delta = \frac{f_u - f_d}{S_0(u-d)}$, we have

$e^{-rT}(\frac{f_u - f_d}{u-d} \times u - f_u) = \frac{f_u - f_d}{u-d} - f$

$e^{-rT}\left(\frac{u \times f_u - u \times f_d - u \times f_u + d \times f_u}{u-d}\right) = \frac{f_u - f_d}{u-d} - f$

$e^{-rT}\left(\frac{-u \times f_d + d \times f_u}{u-d}\right) = \frac{f_u - f_d}{u-d} - f$

$f = \frac{f_u - f_d - e^{-rT}(f_u \times d - u \times f_d)}{u-d}$

$f = e^{-rT}\left(\frac{e^{rT}(f_u - f_d) - f_u \times d + u \times f_d}{u-d}\right)$

$f = e^{-rT}(\frac{(e^{rT} - d)f_u}{u-d} + \frac{(u - e^{rT})f_d}{u-d})$

Note $1 - \frac{e^{rT} - d}{u-d} = \frac{u - d - e^{rT} + d}{u-d} = \frac{u - e^{rT}}{u-d}$

$f = e^{-rT}(p \times f_u + (1-p) \times f_d)\ where\ p = \frac{e^{rT} - d}{u-d}$

2. A stock price is currently \$50. In each of the next two three-month periods, the stock price has an equal probability of increasing by 6% or decreasing by 5%. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a six-month European call option with a strike price of \$51? Check your answer using the simulator.

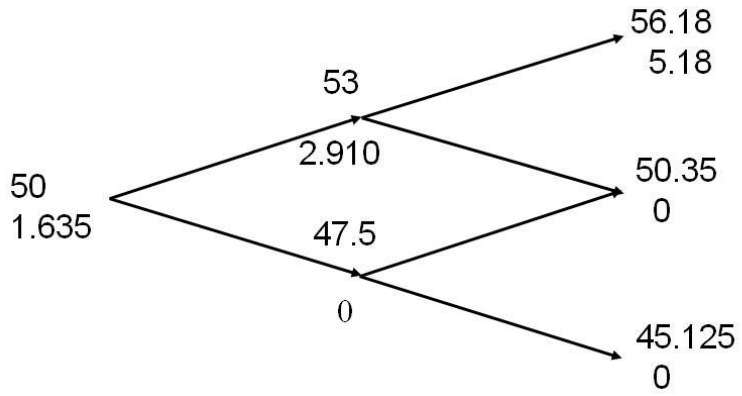
A tree describing the behavior of the stock price is shown below. The risk-neutral probability of an up move, p, is given by

$$p = \frac{e^{0.05\times 3/12} - 0.95}{1.06 - 0.95} = 0.5689$$

There is a payoff from the option of $56.18 - 51 = 5.18$ for the highest final node (which corresponds to two up moves) and zero in all other cases. The value of the option is therefore

$$5.18 \times 0.5689^2 \times e^{-0.05\times 6/12} = 1.635$$

This can also be calculated by working back through the tree. The value of the call option is the lower number at each node in the figure.



3. For the situation considered in Question 2, what is the value of a six-month European put option with a strike price of \$51? Verify that the European call and European put prices satisfy put–call parity. Check your answer using the simulator.

The tree for valuing the put option is shown in Figure below. We get a payoff of $51 - 50.35 = 0.65$ if the middle final node is reached and a payoff of $51 - 45.125 = 5.875$ if the lowest final node is reached. The value of the option is therefore

$$(0.65 \times 2 \times 0.5689 \times 0.4311 + 5.875 \times 0.4311^2)e^{-0.05\times 6/12} = 1.376$$

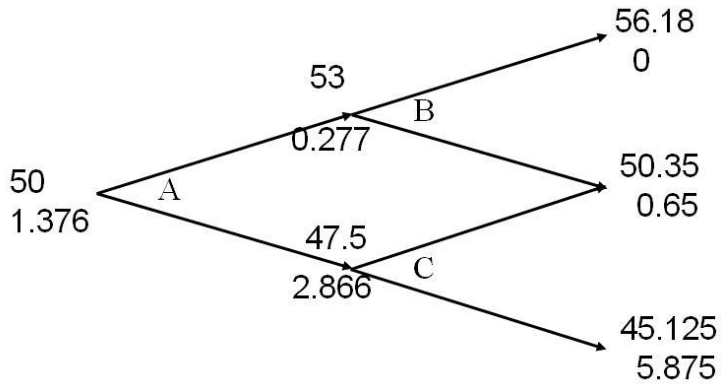
This can also be calculated by working back through the tree.
The value of the put plus the stock price is

$$1.376 + 50 = 51.376$$

The value of the call plus the present value of the strike price is

$$1.635 + 51e^{-0.05\times 6/12} = 51.376$$

This verifies that put–call parity holds.



4. How do the answers to question 2 and 3 change if the probability of an up-move increases to 55% but all other details remain the same?

The option price does not change if only the real-world probability of an up-move increases, as long as the current stock price, up and down factors (*u* and *d*), risk-free rate, and other inputs remain unchanged.
This is because option pricing in the binomial model uses risk-neutral probabilities, not real-world probabilities. These depend only on the risk-free rate, *u*, and *d*; not on how likely an up-move is in reality.
However, in reality, the current spot price of the stock is itself influenced by investors' beliefs about the real-world probabilities of future price movements. For example, if investors believe the stock is more likely to go up (higher physical probability of an up-move), they may be willing to pay more for it *today*, pushing the current spot price higher. Since option values depend on the current stock price, this could indirectly affect the option price. But within the model, with all inputs fixed, the physical probability has no effect.

5. A stock price is currently \$50. It is known that at the end of two months it will be either \$53 or \$48. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a two-month European call option with a strike price of \$49? Use no-arbitrage arguments.

At the end of two months the value of the option will be either \$4 (if the stock price is \$53) or \$0 (if the stock price is \$48). Consider a portfolio consisting of:

$$+\Delta \quad : \quad \text{shares}$$

$$-1 \quad : \quad \text{option}$$

The value of the portfolio is either 48Δ or $53\Delta - 4$ in two months. If

$$48\Delta = 53\Delta - 4$$

i.e.,

$$\Delta = 0.8$$

the value of the portfolio is certain to be 38.4. For this value of Δ the portfolio is therefore riskless. The current value of the portfolio is:

$$0.8 \times 50 - f$$

where f is the value of the option. Since the portfolio must earn the risk-free rate of interest

$$(0.8 \times 50 - f)e^{0.10 \times 2/12} = 38.4$$

i.e.,

$$f = 2.23$$

The value of the option is therefore \$2.23.
This can also be calculated directly from equations (12.2) and (12.3). $u = 1.06$, $d = 0.96$ so that

$$p = \frac{e^{0.10\times 2/12} - 0.96}{1.06 - 0.96} = 0.5681$$

and

$$f = e^{-0.10\times 2/12} \times 0.5681 \times 4 = 2.23$$